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Distinguished National Professor
Director, Natural Product Research Unit
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Date and Place of Birth : April 4, 1955, Faisalabad (Pakistan)

Educational Background:

1985 **PhD**–Pharmacology, University of Sydney, Australia
1978 **MSc** (Hon.)–Physiology & Pharmacology, University of Agriculture, Faisalabad, Pakistan

Professional Background:

1979-86 Lecturer, Dept. Pharmacy, Bahauddin Zakariya University, Multan, Pakistan
1986-97 Senior Instructor, Assistant Professor and Associate Professor, Dept. Pharmacology, Aga Khan University Medical College (AKUMC), Karachi, Pakistan
1997-date Professor, Department of Physiology and Pharmacology, AKUMC, Karachi, Pakistan
2004-date Distinguished National Professor, Title and research grant by the Higher Education Commission of Pakistan
1995-date Adjunct Professor, Faculty of Pharmacy and HEJ Res Institute of Chemistry, University of Karachi, Pakistan
2010- Visiting Professor, King Saud University, Riyadh, Saudi Arabia

Academic leadership

2001-2003 **Deputy Chair**, Department of Biological and Biomedical Sciences (Rotatory position amongst senior faculty of the Department)
2002-date **Founding Director**, Natural Products Research Unit, DBBS
www.aku.edu/medicalcollege/bbs/natprod.htm
2008-date **Vice Chair** for Research and Graduate Studies, DBBS, AKU, Karachi
2008-date **Programme Director**, PhD in Health Sciences, AKU
2008-2009 **Interim Chair**, Board of Graduate Studies, Aga Khan University
2009-date **President**, Pakistan Chapter, International Society of Pharmacoeconomics and Operational Research (ISPOR)
http://www.ispor.org/regional_chapters/Pakistan/index.asp

Academic Honors/Awards

- Life-time Academic Achievement Award, by the Higher Education Commission, Government of Pakistan 2011.
- Member, WHO Expert Committee on Selection & Use of Essential Medicine (2006-date).
- Member, Executive/Advisory Committee, International Society of Pharmacoeconomics and Operation Research (Asia Consortium)

- Two Civil Awards by the President of Pakistan (*Pride of Performance and Sitara-i-Imtiaz*)
- Distinguished National Professor, in the area of Pharmaceutical Sciences, by the Higher Education Commission of Pakistan (2004-), Title along with grants
- Occupied 1st National ranking in the fields of Pharmaceutical Sciences (highest total of International Citations/Impact Factor as published by Pakistan Council for Science Technology)
- Invited speaker and chaired scientific sessions at several international symposia.
- Fellow, *Pakistan Academy of Sciences* (elected in 2002)
- Board member, *International Society of Ethnopharmacology* (2002-date)
- WHO Consultant on *Traditional Medicine* (2002-date)
- *Agha Hassan Abedi Medal* in Medical Sciences, Pakistan Academy of Sciences (2000)
- *National Book Foundation Research Award*: 1st prize in Medical Sciences (1998)
- *Fellow (since 1997), Secretary General (1999-2002), Scientific Secretary (2003-date) of Pakistan Academy of Medical Sciences*
- *Outstanding Teacher Award*: Aga Khan University (1996)
- *Ibn-al-Hayatham Prize in Science*, Pakistan Academy of Sciences (1993)
- *Salam Prize in Science* (Biology-1992)- TWAS, Italy.
- *Boehringer International Research Award* (1988): Boehringer Ingelheim Zentr, Germany
- *Sydney University Postgraduate Merit Scholarship*, Australia (1981-85)

Member, Editorial Board:

Fitoterapia (1999-2010), and *Journal of Ethnopharmacology* (2000-) Elsevier Science, *Journal of Natural Remedies* (2002), *Phytotherapy Research*, John Wiley, UK (2006-), *Pharmacognosy Reviews* (2007-), *The Open Bioactive Compounds Journal* (2008-); *The Open Complementary Medicine Journal* (2008-), Bentham Science Publishers, *Journal of Animal and Plant Sciences* (2000-), *Pakistan Journal of Medical Research* (2009-), *Southern Medical News* (2009-), *Journal of Basic and Applied Sciences* (2009-), *World Journal of Gastrointestinal, Pharmacology and Therapeutics* (2010-), *The Natural Products Journal*, Bentham Science Publishers (2010-), *International Journal of Agriculture and Biology* (1999-), *Evidence Based Complementary and Alternative Medicine* (2012-)

Advisor to:

Pakistan Science Foundation (1994-), Pakistan Medical Research Council (1995-), Higher Education Commission (2003-), Ministry of Health (2000-2003), Ministry of Science and Technology (2002-2005), Pakistan Council for Science and Technology (2005-), National Testing Service (2006-), Member, Selection Boards/Board of Studies at different national Universities

WHO Geneva (2000-), World Innovation Foundation (2006-), International Foundation for Sciences (2006-), South African Medical Research Council (2006-), University of Malaya,

Funding:

Aga Khan University, Pakistan Science Foundation, Higher Education Commission of Pakistan, Pakistan Medical Research Council, International Foundation for Science, Sweden.
Secured several travel grants for invited/plenary lecture/seminars at international meetings.

Supervision of Graduate and Elective Students

- 16 PhDs as main supervisor and 6 PhDs as co-supervisor (6 ongoing as sole supervisor)
- 16 Ms/M.Phil as main supervisor and 15 M.Phil as co-supervisor
- Supervise around 15 summer elective students each year from all over the country

Research

- Three patents and co-editor of two Conference Proceedings

- Co-editor of two WHO reports series on Essential medicine
- Published around 300 research papers mostly in journals of international repute with a **cumulative impact factor of over 600**.
- Published around 170 papers in last 6 years (2005-date) in 55 different ISI-indexed journals with cumulative impact factor around 300.
- Six papers of our group ranked amongst the top 25 articles (Survey by ScienceDirect)
- H-Index of 35 based on Google Scholar
- Participated in over 65 international conferences (Invited/plenary speaker at 30)
- Member, Scientific Committee and Chair/Co-Chair of Scientific sessions of several national/international symposia/conferences

Areas of research interest (see details: www.aku.edu/medicalcollege/bbs/natprod.htm)

- **Natural products research** (a wide range of biological activities, such as *antihypertensive, cardiotonic, antidyslipidemic, anti-oxidant, hepatoprotective, antiinflammatory, antidiabetic, anti-urolithic, uterotonic, laxative, prokinetic, antispasmodic, antidiarrhoeal, antiamoebic, antifungal, and Anti-Alzheimer's disease* **with focus on medicinal value of food** and providing scientific basis for the presence of novel combination of activities with synergistic and/or side effects neutralizing potential in medicinal plants.
- **Complementary medicine:** with focus on modifying life style and dietary habits in line with the Sunnah of Holy prophet (prophetic medicine); delivered lectures at different forums
- **Muscarinic Receptor Subtypes** (pioneer work)
- **Public Health and Community Pharmacy**
- **Clinical Studies on herbal products**

Research Collaboration: A network of collaboration established which include at least 8 research groups at national and 6 at international level.

Teaching: Wide experience of teaching pharmacology to medical, pharmacy and nursing students at undergraduate level and designed courses for graduate students and currently heading PhD programme in Health Sciences.

Innovating Teaching: Designed a teaching course for the undergraduate medical students of the Aga Khan University Medical College (Teaching rational prescribing to medical students through critical evaluation of drug promotional materials), which is a self learning exercise with focus on efficacy, safety, patient compliance and cost-effective treatment.

Invited/Plenary Speaker at International Forums in recent years (2006-date)

- **International Symposium on Agriculture in the 21st century**, Centenary Celebrations, University of Agriculture, Faisalabad, March 16-17, 2006.
Title - *Medicinal value of foods and dietary supplements*
- **9th International Congress of Ethnopharmacology**, Nanning, China, August 22-26, 2006 (Plenary Speaker, member of Scientific Committee and chaired a scientific session),
Title- *Novel combinations in medicinal plants with synergistic and side-effects neutralizing potential*
- **12th Asian Symposium on Medicinal Plants and Spice**, Nov 13-18, 2006, Padang, Indonesia.
Title *Novel developments in natural products research* (Invited speaker; chaired two scientific sessions)
- **2nd International Symposium on Cardiovascular risk factors: Pathophysiological aspects and phytopharmacological approaches**. University Sidi Mohamed Ben Abdallah Faculty of Sciences, Fez, Morocco, November 23-25, 2006 (member of Organizing Committee)
Title - *Role of Ethnopharmacology in Modern Medicine* (Plenary speaker)

- Title (*lecture # 2*) *Medicinal value of food and dietary supplements in cardiovascular diseases*
- **World Health Updates**, January 26-28, 2007 Expo Centre, Karachi;
Title: *Integration of traditional and modern medicine for better healthcare in developing countries (plenary lecture)*
 - **12th Diabetes & Endocrine Conference**, March 1-4, 2007, Dhaka, Bangladesh.
Title-*Medicinal value of food and dietary supplements (Plenary Lecture)*
 - **Attended 15th meeting of the WHO Expert Committee on the Selection and Use of Essential Medicine**, March 18-23, WHO Office, Geneva (Finalized the WHO Essential list of Medicine)
 - **3rd International Congress on Traditional Medicine and Materia Medica**, July 17-20, Kuala Lumpur, Malaysia Title: *Novel developments in natural products (Plenary lecture)*.
 - **1st Regional Conference on Traditional Arabic and Islamic Medicine**, August 8-10, Amman, Jordan Title: *Novel developments in medicinal plant research (plenary speaker)*;
 - **18th Iranian congress of physiology and pharmacology**, August 26-30, 2007, Mashhad, Iran.
Title: *Role of ethnopharmacology in modern medicine (plenary lecture)*
 - **International Conference on Traditional Medicine and Medicinal Plants**, September 7-9, 2007 Surabaya, Indonesia. (Plenary speaker) Title: *Obesity control with traditional diet and life style (Plenary lecture)*
 - **16th Meeting of the WHO Expert Committee on the Selection and Use of Essential Medicine**, October 24-25, 2007, WHO Office, Geneva
 - **7th Annual Conference of the Society of Pharmacovigilance**, 23-25 Nov, 2007 Jaipur, India (SoPI) Title: *Safety issues of herbal products*.
 - **International Conference of Plant Scientists**, University of Agriculture, Faisalabad, Pakistan April 21-24, 2008, Plenary speaker and Chaired a Plenary session
Titles- *Role of ethnobotany in the development of modern medicine*
 - **Joint meeting of 10th International Congress of Ethnopharmacology and 20th Symposium of Medicinal Plants of Brazil**, São Paulo, Brazil, Sept 16-19, 2008
Topic- Controlling obesity and metabolic disorders with traditional diet and life style
 - **Scientific Research Outlook and Technology Development in the Arab World**, Fez, Morocco 27-31 October, 2008
Titles: *Integration of traditional and modern medicine for better healthcare (keynote lecture)* *Novel developments in natural products pharmacology*
 - **International Family Medicine Conference**, Karachi, October 30-November 02, 2008
Title: *Healing power of our food and dietary supplements*
 - **Joint meeting of the 1st SAARC Conference on Physiological Societies and 11th Biennial Conference of Pakistan Physiological Society**, Nov 10-12, Islamabad
Title- *Controlling obesity and restoring health with life style and dietary modifications*
 - **Scientific Research Outlook and Technology Development in the Arab World**, Fez, Morocco 27-31 October, 2008 Title: *Integration of traditional and modern medicine for better healthcare* *Novel developments in natural products pharmacology*
 - **International Symposium on Modernization of Traditional Chinese Medicine: Acupuncture and herbal medicine research**, May 14-16, 2009, Graz, Austria.
Title: *Synergistic and/or side-effects neutralizing combinations in herbal medicine*
 - **1st World Conference on Pharmacology of Natural and Traditional Medicine**, Hangzhou, China. Sept 9-12, 2009. Topic: *New leads towards drug discovery from natural products*
 - **Workshop on Vital role of preclinical studies in Safety and effectiveness of drugs**, Oct 25-29, 2009, King Fahad Medical Research Center , King Abdul-Aziz University, Jeddah, Saudi Arabia
Titles: *Global Trends in the Use of Traditional Medicine and WHO Traditional Medicine Strategy*
Role of natural products in modern medicine
Prophetic medicine: a way towards a healthy life and prosperous society
Controlling obesity and metabolic disorders with dietary and life style modifications

- **Seminar on Clinical Trials and Pharmacovigilance Awareness and Current Trends in Pakistan**, Jointly organized by the Ministry of Health and Dow University of Health Sciences, Karachi, May 24, 2010
Title: Safety issues (Pharmacovigilance) in using medicinal products
- **Pakistan Academy of Sciences, Islamabad, June 14, 2010**
Title: Preventing illness and maintaining health with life style changes and natural foods
- **6th World Congress of Basic and Clinical Pharmacology**, Copenhagen, 17-23 July, 2010, organized by the International Union of Pharmacologists. <http://www.worldpharma2010.org/>.
Title: Novel synergistic and/or side-effects neutralizing combinations in natural products
- **The 12th International Symposium on Natural Products Chemistry (ISNPC-12)**, November 22- 25, 2010, Karachi (also chaired a session)
Title: Novel development in natural products research
- **3rd International Symposium-cum Training Course on Molecular Medicine and Drug Research**, January 2-6, 2011, Karachi (Prof. Sheikh Arshad Saeed memorial plenary lecture)
Title: Trends in Ethnopharmacology
- **16th Annual meeting of the International Society of Pharmacoeconomics and Outcome Research**, May 21-25, Hilton Baltimore, US (Travel grant \$ 2,500/= as an invited speaker). www.ispor.org
Title: Lead detoxifying effect of vitamin c in traffic police subjects
Progress and activities of Pakistan Chapter (Action meeting)
- **The 12th International Symposium on Natural Products Chemistry (ISNPC-12)**, November 22 - 25, 2010, Karachi (also Chaired a session)
Title: Novel development in natural products research
- **3rd International Symposium-cum Training Course on Molecular Medicine and Drug Research**, January 2-6, 2011, Karachi (plenary speaker)
Title: Trends in Ethnopharmacology
- **King Saud University, College of Pharmacy, Riyadh, Saudi Arabia, March 3-16, 2011**
Title: Controlling obesity and cardiovascular diseases with life style and dietary modification
- **Pakistan Council for Science and Technology**, Islamabad, January 12, 2010
Title: Controlling blood pressure and other cardiovascular diseases with nutrition and life style approaches
- **Distinguished lecture series, Department of Biochemistry, University of Karachi, March 1st, 2012**, Title: *Controlling obesity and restoring good health with life style and dietary modifications*
- **26th Meeting of Malaysian Society of Pharmacology and Physiology**, May 18-20, Penang, Malaysia. Title: *Controlling obesity and restoring physiological functions with life style and dietary modifications*

RECENT RESEARCH PUBLICATIONS (2005-2012)

Research articles: over 160 in 50 different journals with Cumulative Impact factor around 300
Review articles: 08, Book Chapters: 07, WHO Reports series: 02

WHO Expert Committee Reports Series

1. Bero L, Cranswick N, Fernandopulle R, **Gilani AH**, et al. (2007). The selection and use of essential medicines: 15th model list of essential medicines. **World Health Organization Technical Report Series**. (946): back cover, v-162.
2. Al-Fannah JMA, Coelho H, Cranswick N, **Gilani AH**, et al. (2008). The selection and use of essential medicines: model list of essential medicines for children. **World Health Organization Technical Report Series**. (950): back cover, vii-174.

Research Papers

3. Choudhary MI, Nawaz SA, Zaheer-ul-Haq, Azim MK, Ghayur MN, Lodhi MA, Jalil S, Khalid A, Ahmed A, Rode , Atta-ur-Rahman, **Gilani AH**, Ahmad VU (2005). Juliflorine: a potent natural peripheral anionic-site- binding inhibitor of acetylcholinesterase with calcium-channel blocking potential, **a leading candidate for Alzheimer's disease** therapy. *Biochemical and Biophysical Research Communications* 332: 1171-1179.
4. **Gilani AH**, Shah AJ, Ghayur MN, Majeed K (2005). Pharmacological basis for the use of turmeric in gastrointestinal and respiratory disorders. *Life Sciences* 76: 3089-3105.
5. Ghayur MN and **Gilani AH** (2005) Pharmacological basis for the medicinal use of Ginger in gastrointestinal disorders. *Digestive Diseases and Sciences* 50: 1889-1897.
6. Choudhary MI, Nawaz SA, Zaheer-ul-Haq, Lodhi MA, **Ghayur MN**, Jalil S, Riaz N, Yousuf S, Malik A, Rode BM..... **Gilani AH**, Atta-ur-Rahman (2005). Withanolides, a new class of natural cholinesterase inhibitors with calcium antagonistic properties. *Biochemical and Biophysical Research Communications* 334: 276-287.
7. Ghayur, MN, **Gilani, AH** (2005) Ginger lowers blood pressure through blockade of voltage-dependent calcium channels. *Journal of Cardiovascular Pharmacology* 45, 74-80.
8. **Gilani AH**, Khan A, Subhan F, Khan M. (2005) Antispasmodic and bronchodilator activities of St. John's Wort are putatively mediated through dual inhibition of calcium influx and phosphodiesterase. *Fundamental and Clinical Pharmacology* 19: 695-705.
9. Rohra DK, Sharif H, Zuberi H, Ghayur MN, **Gilani AH** (2005) Acidosis-induced relaxation of human mammary artery is due to activation of ATP-sensitive potassium channels. *European Journal of Pharmacology* 514: 175-181
10. Ghayur MN, **Gilani AH**, Houghton P (2005) Species differences in the gut stimulatory effects of radish seeds. *J Pharmacy and Pharmacology* 57: 1493-1501.
11. **Gilani, AH**, Khan A, Shah AJ, Connor J and Jabeen Q (2005) Blood pressure lowering effect of olives is mediated through calcium channel blockade. *International Journal of Food Sciences and Nutrition* 56: 613-620.
12. Butt ZA, **Gilani, AH**, Nanani, D, Sheikh AL, White, F (2005). Quality of Pharmacies in Pakistan: A Cross-sectional Survey. *Int Journal for Quality in Health Care* 17:307-313
13. Ghayur MN, **Gilani AH**, Afridi MB, Houghton PJ (2005). Cardiovascular effects of ginger aqueous extract and its phenolic constituents are mediated through multiple pathways. *Vascular Pharmacology* 43(4): 234-241.
14. **Gilani AH**, Bashir S, Janbaz KH, & Khan, A. (2005) Pharmacological basis for the use of *Fumaria indica* in constipation and diarrhea. *Journal of Ethnopharmacology* 96: 585-589.
15. **Gilani AH**, Ghayur, MN, Khalid, A, Zaheer-ul-Haq, Choudhary, MI, Atta-ur-Rahman 2005. Presence of antispasmodic, antidiarrhoeal, antisecretory, calcium antagonist and acetylcholinesterase inhibitory steroidal alkaloids in *Sarcococca saligna*. *Planta Medica* 71: 120-125.
16. **Gilani, AH**, Khan A, Shah AJ, Jabeen Q, Subhan, F. and Ghafar, R Q (2005). Antispasmodic and blood pressure lowering effects of *Valeriana wallichii* are mediated through K⁺ channel activation. *Journal of Ethnopharmacology* 100(3): 347-352.
17. **Gilani, A.H.**, Jabeen Q, Ghayur, M.N. Janbaz, K.H, Akhtar, M.S. (2005) Studies on the antihypertensive, antispasmodic, bronchodilator and hepatoprotective activities of the *Carum copticum* seed extract. *Journal of Ethnopharmacology* 98:127-135.
18. **Gilani, AH** (2005). Role of ethnopharmacology in the development of modern medicine. In: Shinwar ZK, Watanabe T, Ali M, Anwar R (Eds.) Proceedings of the *International Symposium on Medicinal Plants: Linkages beyond national boundaries*. Al-Aziz Communications, Peshawar, pp. 192-197.
19. **Gilani AH** (2005). Role of medicinal plants in modern medicine. *Malaysian Journal of Science* 24: 1-5 (review article).

20. Amor EC, Villaseñora IM, Ghayur MN, **Gilani AH**, Choudhary MI (2005). Spasmolytic flavonoids from *Syzygium samarangense* (Blume) Merr. & L.M. Perry. *Zeitschrift für Naturforschung c* 60: 67-71.
21. **Gilani AH**, Bukhari I, Khan, RA, Farman-ullah Ahmad VU (2005) Cholinomimetic and calcium channel blocking activities of *Carthamus oxycanth*. *Phytotherapy Research* 19:679-683.
22. **Gilani AH**, Bashir S, Janbaz KH, Shah AJ (2005). Presence of cholinergic and calcium channel blocking activities explains the traditional use of *Hibiscus rosasinensis* in constipation and diarrhea. *Journal of Ethnopharmacology* 102: 289-294.
23. Ghayur MN, **Gilani AH** (2005) Gastrointestinal stimulatory and uterotonic activities of dietary radish leaves extract are mediated through multiple pathways. *Phytotherapy Res* 19:750-75
24. **Gilani, AH**, Yaeesh, S, Jamal, Q, Ghayur MN (2005) Hepatoprotective activity of aqueous-methanol extract of *Artemisia vulgaris*. *Phytotherapy Research* 19: 170-172
25. **Gilani AH**, Khan A, Ghayur MN, Herzig JW (2006) Antispasmodic effects of Rooibos tea (*Aspalathus linearis*) are mediated predominantly through potassium channel activation. *Basic and Clinical Pharmacology and Toxicology* (formerly *Pharmacol. Toxicol*) 99:365-373.
26. Khan A, **Gilani AH** (2006). Selective bronchodilatory effect of Rooibos tea (*Aspalathus linearis*) and its flavonoid, chrysoeriol. *European Journal of Nutrition* 45: 463-469
27. Taqvi SIH, Ghayur MN, **Gilani AH**, Saify ZS, Aftab MT (2006). Synthesis and smooth muscle-selective relaxant activity of a piperidine analogue: 1-(4'-fluorophenacyl)-4-hydroxy-4-phenylpiperidinium chloride. *Archives of Pharmacal Research* 29: 34-39.
28. Iqbal Z, Lateef M, Jabbar A, Ghayur, MN, and **Gilani AH** (2006) In vitro and in vivo anthelmintic activity of *Nicotiana tabacum* L. leaves against gastrointestinal nematodes of sheep. *Phytotherapy Research* 20: 46-48.
29. Sabira Begum, Syed Imran Hassan[†], Bina S. Siddiqui[†], Rehana Ifzal, Sobiya Perwaiz, Tahira Kiran, Farhana Shaheen, M. Nabeel Ghayur and **Gilani AH** (2006) Preparation, Structure and Spasmolytic Activity of Some Derivatives of Harmine Series of Alkaloids. *Natural Products Research* 20: 213-227.
30. Ghayur MN, **Gilani AH** (2006). Radish seed extract mediates its cardiovascular inhibitory effects via muscarinic receptor activation. *Fundamental and Clinical Pharmacology* 20:57-63
31. **Gilani AH**, Ghayur MN, Houghton PJ, Jabeen Q, Kazim SF, Juman MI, Saeed SA (2006). Studies on the hypotensive, cardio-suppressant, vasodilator and antiplatelet activities of betel nut crude extract and its constituents. *International Journal of Pharmacology* 2:33-41
32. Ghayur MN, **Gilani AH**, Khan A, Amor EC, Villaseñor IM, Choudhary MI (2006). Presence of calcium antagonist activity explains the use of *Syzygium samarangense* in diarrhea. *Phytotherapy Research* 20: 49-52.
33. Taqvi SIH, Aftab MT, Ghayur MN, **Gilani AH**, Saify ZS (2006). Synthesis and biological evaluation of 1-(2', 4'-dimethoxyphenacyl)-4-acetyl-4-phenylpiperidinium bromide in intestinal and cardiovascular tissues. *Journal of Pharmacology and Toxicology* 1: 126-133.
34. Rohra DK, **Gilani AH**, Memon IK, Perven G, Khan MT, Zafar H, Ahuja RK (2006) Critical evaluation of the claims made by pharmaceutical companies in drug promotional material in Pakistan. *Journal of Pharmacy and Pharmaceutical Sciences* 9: 50-59.
35. Taqvi SIH, Aftab MT, Ghayur MN, **Gilani AH**, Saify ZS (2006). Synthesis and pharmacological screening of 1-(2', 4'-dimethoxyphenacyl)-4-hydroxy-4-phenylpiperidinium bromide. *International Journal of Pharmacology* 2: 146-151.
36. Taqvi SIH, Ghayur MN, **Gilani AH**, Aftab MT, Saify ZS (2006). Synthesis, spasmolytic, cardio-suppressant, vasodilator and Ca⁺⁺ antagonist activities of 1-[4'-methylphenacyl]-4-acetyl-4-phenylpiperidinium bromide. *Pakistan Journal of Biological Sciences* 9: 60-66.
37. Iqbal Z, Lateef M, Jabbar A, Ghayur, MN, and **Gilani AH** (2006) In vivo anthelmintic activity of *Butea monosperma* against trichostrongylid nematodes in sheep. *Fitoterapia* 77: 137-140.
38. Iqbal Z, Lateef M, Akhtar MS, Ghayur MN, **Gilani AH** (2006) In vivo anthelmintic activity of ginger against gastrointestinal nematodes of sheep. *Journal of Ethnopharmacology* 106:285-287

39. **Gilani, AH**, Khan A, Ghayur, MN (2006) Calcium antagonist and cholinomimetic activities explains the medicinal uses of olives in gut disorders. **Nutrition Research** 26: 277-283.
40. Yaeesh S, Jamal Q, Khan A, **Gilani AH** (2006) Studies on hepatoprotective, antispasmodic and calcium antagonist activities of the aqueous-methanol extract of *Achillea millefolium*. **Phytotherapy Research** 20: 546-551.
41. Ghayur, MN, **Gilani AH** (2006). Species differences in the prokinetic effects of ginger. **International Journal of Food Sciences and Nutrition** 57: 65-73.
42. Bashir S, Janbaz KH, Jabeen Q and **Gilani AH** (2006) Studies on spasmogenic and spasmolytic activities of *Calendula officinalis* flowers. **Phytotherapy Research** 20: 906-910.
43. Lateef M, Iqbal Z, Iqbal Z, Jabbar A, Akhtar MS, MN Khan, **Gilani AH** (2006) Preliminary screening of *Trachyspermum ammi* (L.) seed for anthelmintic activity in sheep. **Tropical Animal Health and Production** 38: 491-96.
44. Ghayur MN, **Gilani AH** 2006. Studies on cardio-suppressant, vasodilator and tracheal relaxant effects of *Sarcococca saligna*. **Archives of Pharmacal Research** 29: 990-2006.
45. **Gilani AH**, Shah AJ, Manzoor Ahmad M, Shaheen F, (2006.) Antispasmodic effect of *Acorus calamus* is mediated through calcium channel blockade. **Phytotherapy Res** 20: 1080-1084.
46. Atta-ur-Rahman, Khalid A, Sultana N, Ghayur M N, Mesaik M A, Khan M R, **Gilani AH** Choudhary M I 2006. New Natural Cholinesterase Inhibiting and Calcium Channels Blocking Quinoline Alkaloids, **Journal of Enzyme Inhibition and Medicinal Chemistry** 21: 703-710.
47. Khan H, Ghayur MN, **Gilani AH**. (2006). *In vitro* studies on antispasmodic and vasodilator activities of (+)-Catechin, a commonly occurring flavonoid. **European Journal of Medical Research** 11(Supplement II): 125-126.
48. Ghayur MN, **Gilani AH** (2007) Inhibitory activity of ginger rhizome on airway and uterine smooth muscle preparations **European Food Research and Technology** 224: 477-481.
49. Mustafa MR, Ajay M, Chai HJ, Mustafa AM, **Gilani AH**. 2007. Mechanisms of the anti-hypertensive effect of *Hibiscus sabdariffa* L. calyces **Journal of Ethnopharmacology** 109:388-393
50. Ghayur MN, **Gilani AH** (2007) Cardio-selective inhibitory effect of the betel nut extract: possible explanation. **Pharmazie** 62: 67-71.
51. Subhan F, Khan M, **Gilani AH** 2007. Antagonism of antinociceptive effect of hydro-ethanolic extract of *Hypericum perforatum* Linn. by a non-selective opioids receptor antagonist, naloxone. **Pakistan Journal of Biological Sciences** 10: 792-796
52. Ghayur, MN, **Gilani AH** 2007. α -adrenoceptor mediated hypertensive and vasoconstrictor effects of dietary radish leaves extract. **Journal of Health Sciences** 53: 151-155.
53. Jabeen Q, Aziz N, **Gilani AH**, (2007). The spasmogenic and spasmolytic activities of *Lavandula stoechas* are mediated through muscarinic receptor stimulation and calcium channel blockade. **International Journal of Pharmacology** 3: 61-67.
54. Bukhari IA, Khan RA, **Gilani AH**, Shah AJ, Hussain J, Ahmad VU. 2007. The analgesic, anti-inflammatory and calcium antagonist potential of *Tanacetum artemisioides*. **Archives of Pharmacal Research** 30: 303-312.
55. Ali A, **Gilani AH** (2007) Medicinal value of ginger with focus on its use in nausea and vomiting of pregnancy. **International Journal of Food Properties** 10: 269-278 (review).
56. Ghayur MN, Khan AH, **Gilani AH**. 2007. Ginger facilitates cholinergic activity possibly due to blockade of muscarinic autoreceptors in rat stomach fundus. **Pakistan Journal of Pharmaceutical Sciences** 20(3):231-5.
57. **Gilani AH**, Shah AJ, Yaeesh S. (2007) Presence of cholinergic and calcium antagonist constituents in *Saussurea lappa* explains its use in constipation and spasm. **Phytotherapy Research** 21: 541-544.
58. Nisar M, Adzu B, Inamullah K, Bashir A, Ihsan A, **Gilani AH** (2007) Antinociceptive and antipyretic activities of the *Zizyphus oxyphylla* Edgew. Leaves **Phytotherapy Res** 21: 693-695.
59. Ghayur MN, Khan H, **Gilani AH** (2007) Antispasmodic, Bronchodilator and Vasodilator Activities of (+)-Catechin, a Naturally Occurring Flavonoid. **Arch Pharmacol Res** 30:970-975.

60. **Gilani AH**, Shah AJ, Janbaz KH, Ahmed SP, Ghayur MN (2007) Studies on antihypertensive and antispasmodic activities of *Andropogon muricatus* Nash. **Canadian Journal of Physiology and Pharmacology** 85 (9): 911-917.
61. **Gilani AH**, Bashir S, Khan A (2007) Pharmacological basis for the use of *Borago officinalis* in gastrointestinal, respiratory and cardiovascular disorders. **J Ethnopharmacology** 114: 393-399.
62. Ullah F, Hussain H, Hussain J, Bukhari IA, Khan MT, Choudhary MI, **Gilani AH**, Ahmad VU (2007) Tyrosinase inhibitory pentacyclic triterpenes and analgesic and spasmolytic activities of methanol extracts of *Rhododendron collettianum*. **Phytotherapy Research**. 21:1076-1081.
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Review Articles:

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